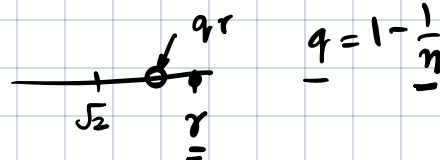


# Functions

sin

Idea:



A function  $f$  is a mapping from a set  $A$  to a set  $B$

$$f: \underbrace{A}_{\text{domain}} \rightarrow \underbrace{B}_{\text{codomain}}$$

Find a rational  $q$  such that  $q < 1$  and  $qr > \sqrt{2}$

For every  $x \in A$ , the mapping  $f(x) \in B$  must be well-defined.

$$\underbrace{\left(1 - \frac{1}{n}\right)r}_{= n} > \sqrt{2}$$

$$f(x) = \pm x$$

$$\left. \begin{array}{l} f(x) = x \\ f(x) = x^2 \\ f(x) = x^3 \end{array} \right\} \text{polynomial}$$

$$f: \mathbb{R} \rightarrow \mathbb{R}$$

$$f(x) = x^{3/2}$$

$$f: \underbrace{\mathbb{R}^+}_{\text{set of all positive real numbers}} \rightarrow \mathbb{R}$$

$$f(x) = x^\alpha$$

$$\left\{ \begin{array}{ll} \alpha \in \mathbb{N} \cup \{0\} & \text{then } \underline{\text{dom}(f)} = \mathbb{R} \\ -\alpha \in \mathbb{N} & \text{then } 0 \notin \text{dom}(f) \\ & \text{dom}(f) = \mathbb{R}^+ \text{ otherwise.} \end{array} \right.$$

$$f(x) = e^x, \text{ exponentials}$$

$$f(x) = \sin x, \cos x$$

$$\rightarrow f(x) = e^{ix} \quad f: \mathbb{R} \rightarrow \mathbb{C}$$

complex exponentials

$$f(x) = \log x$$

$$\text{dom } f = \mathbb{R}^+$$

$$f(x, y) =$$

$\downarrow$   $\searrow$   
 $A_1$   $A_2$

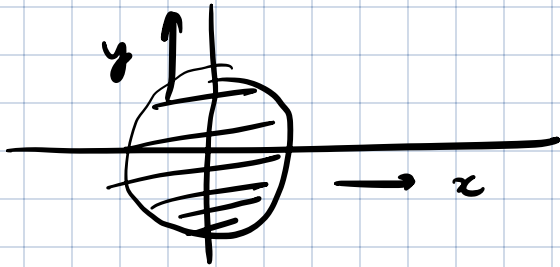
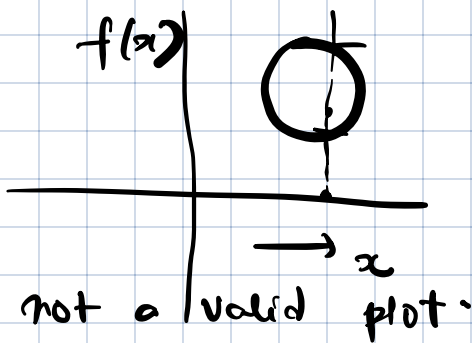
$$\text{dom } f = A_1 \times A_2$$

Cartesian product of sets  $A_1$  &  $A_2$  is the set of all pairs of elements, with the first element of the pair taken from  $A_1$  & the second from  $A_2$

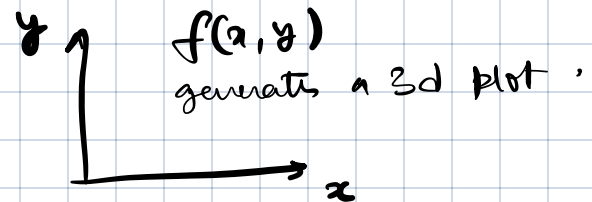
$$A_1 \times A_2 = \{(x, y) \mid x \in A_1, y \in A_2\}$$

$$f(x, y) = \sqrt{1 - x^2 - y^2}$$

$$\text{dom}(f) = \{(x, y) \mid x^2 + y^2 \leq 1\}$$

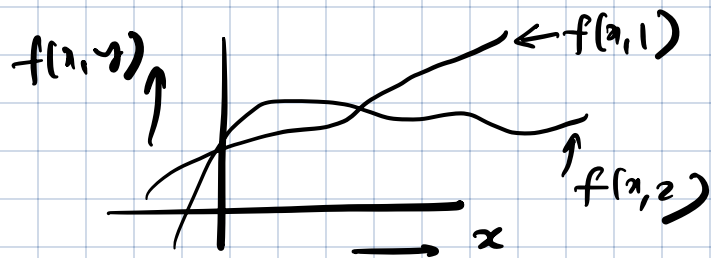
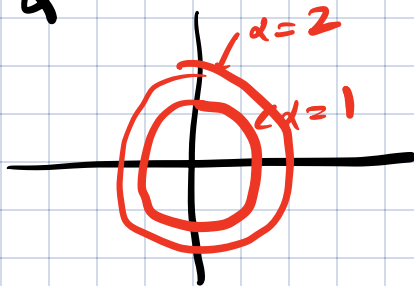


$$f(x, y) = x^2 + y^2$$



level curves

$$f(x, y) = \alpha$$



One-one functions

We say that  $f$  is one-one if

$$f(x) = f(y) \Rightarrow x = y.$$

$$x \neq y \Rightarrow f(x) \neq f(y)$$

$$f(x) \leftarrow \boxed{f} \leftarrow x$$

## Onto functions

$$\forall y \in \text{codom}(f), \exists x \text{ such that } f(x) = y.$$

## Left & right inverses:

We say that a function  $f$  is left-invertible if

$$\exists g \text{ such that } g(f(x)) = x \forall x \in \text{dom} f$$

$g$  is called the left inverse of  $f$ .

Theorem:  $f$  is left-invertible if and only if  $f$  is one-one.

Proof:  $f$  one-one  $\Rightarrow f$  left-invertible  
(Exercise) ↷

$$f \text{ left-invertible} \Rightarrow f \text{ one-one.}$$

Suppose  $f$  is left-invertible and  $x, y$  be such that  $f(x) = f(y)$ .

$$\text{then } \underbrace{g(f(x))}_{=} = \underbrace{g(f(y))}_{=}$$

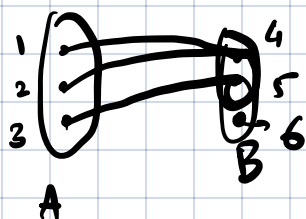
Proposition

Lemma  
↓  
theorem  
contrary

$$f(x) = f(y) \\ x = y$$

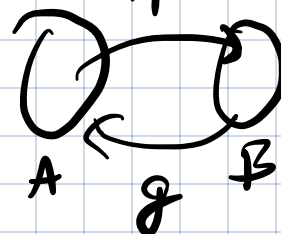
$$x = y$$

we say that  $f$  is right invertible if  $\exists g$  such that  $\underbrace{f(g(y)) = y}_{\forall y \in \text{codom}(f)}$   $g: \text{range}(f) \rightarrow \text{dom}(f)$



$$f(1) = f(2) = 4$$

$$f(3) = 5$$



$$g(4) = 1$$

$$g(5) = 3$$

$$f(g(4)) = f(1) = 4$$

$$f(g(5)) = f(3) = 5$$

Theorem:  $f$  is right invertible if & only if  $f$  is onto.

In general, there can be multiple right & left inverses.

Proof: (I)  $f$  is right invertible  $\Rightarrow f$  is onto  
*Exercise*,

(II)  $f$  is onto  $\Rightarrow f$  is right invertible

(II): If  $f$  is onto, then for any  $y \in \text{codom}(f)$ ,

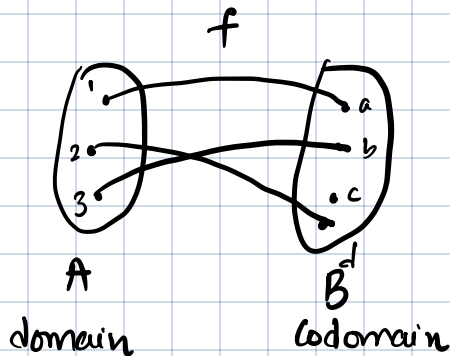
$\exists$  an  $x \in \text{dom}(f)$  such that  $f(x) = y$ .

pick one of these  $x$  & set

$$g(y) := x.$$



Theorem: A function has both left & right inverses if & only if it's one-one & onto. Furthermore, this inverse is unique.



Can I find  $g: B \rightarrow A$  such that  $g(\underbrace{f(x)}) = x$   $\forall x \in A$ .  
left inverse

$$g(a) = 1$$

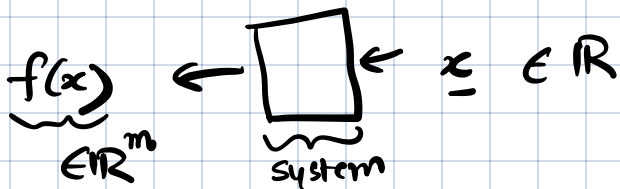
$$g(b) = 3$$

$g(c) = ? \leftarrow$  Any assignment  $g(c) \in \{1, 2, 3\}$

$$g(d) = 2$$

# LINEAR FUNCTIONS

↑ superposition



$$\underline{f(x) = 5x}$$

$$\underbrace{5x_1}_{\text{red}} f(x_1) \leftarrow \square \leftarrow x_1$$

$$\underline{\underline{5x_2}} f(x_2) \leftarrow \square \leftarrow x_2$$

$$f(x_1) + f(x_2) \leftarrow \boxed{\phantom{0}} \leftarrow x_1 + x_2$$

$$S(x_1, x_2)$$

$$= \underline{5x_1} + \underline{5x_2}$$

$$\square \leftarrow x \in \mathbb{R}^n$$

$$\mathbb{R}^n := \mathbb{R} \times \mathbb{R} \times \mathbb{R} \times \dots \times \mathbb{R}$$

elements of  $\mathbb{R}^n$  as "vectors"

$$\underline{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \in \mathbb{R}^n.$$

$$1.5 \begin{pmatrix} 5 \\ 1 \\ 3 \end{pmatrix} = \begin{pmatrix} 7.5 \\ 1.5 \\ 4.5 \end{pmatrix}$$

$$\text{apple} \times \begin{pmatrix} \text{apple} \\ \text{mango} \\ \text{orange} \end{pmatrix} + \begin{pmatrix} \text{banana} \\ \text{watermelon} \end{pmatrix} \quad ?$$

"vector space"

Definition: We say that  $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$  is a linear function if

① SUPERPOSITION  $f(\underline{x}_1 + \underline{x}_2) = f(\underline{x}_1) + f(\underline{x}_2) \quad \forall \underline{x}_1, \underline{x}_2 \in \mathbb{R}^n$

② HOMOGENEITY  $f(\alpha \underline{x}) = \alpha f(\underline{x}) \quad \forall \underline{x} \in \mathbb{R}^n, \alpha \in \mathbb{R}$

E1  $f: \mathbb{R} \rightarrow \mathbb{R} \quad f(x) = 5x$   
is linear.

$$f(x) \leftarrow \boxed{f} \leftarrow x$$

$$\alpha f(x) \leftarrow \boxed{\phantom{f}} \leftarrow \alpha x$$

E2  $f: \mathbb{R}^2 \rightarrow \mathbb{R} \quad f\left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}\right) = 5x_1 + 10x_2$

$$f\left(\begin{pmatrix} y_1 \\ y_2 \end{pmatrix}\right) = 5y_1 + 10y_2$$

$$f\left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}\right) = f\left(\begin{pmatrix} x_1 + y_1 \\ x_2 + y_2 \end{pmatrix}\right)$$

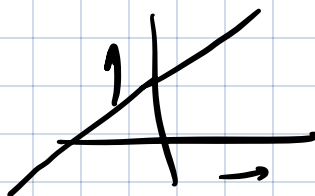
$$= 5(x_1 + y_1) + 10(x_2 + y_2)$$

$$= \underbrace{(5x_1 + 10x_2)}_{f\left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}\right)} + \underbrace{(5y_1 + 10y_2)}_{f\left(\begin{pmatrix} y_1 \\ y_2 \end{pmatrix}\right)}$$

E3:  $f: \mathbb{R} \rightarrow \mathbb{R}$

$$f(x) = 5x + 2$$

$$= f\left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}\right) + f\left(\begin{pmatrix} y_1 \\ y_2 \end{pmatrix}\right)$$



Lemma: Let  $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$  be a linear function, then

$$f\left(\underbrace{\underline{0}}_{\mathbb{R}^n}\right) = \underbrace{\underline{0}}_{\mathbb{R}^m}$$

Proof:  $f$  satisfies Homogeneity

$$\text{Set } \alpha = 0$$

$$f\left(\underbrace{0 \cdot \underline{x}}_{\mathbb{R}^n}\right) = 0 \cdot f(\underline{x})$$

$$f(\underline{0}) = \underline{0}$$

$$f(\alpha \underline{x}) = \alpha f(\underline{x})$$

Linear functions  $f: \mathbb{R} \rightarrow \mathbb{R}$  correspond to lines through the origin.

$f: \mathbb{R}^2 \rightarrow \mathbb{R}$  planes through the origin.

$f: \mathbb{R}^n \rightarrow \mathbb{R}$  hyperplane through origin  
( $n \geq 3$ )

$$f: \mathbb{R} \rightarrow \mathbb{R}$$

$$f(x) = kx \leftarrow \text{direct proportionality}$$

E4:  $f: \mathbb{R}^2 \rightarrow \mathbb{R}^2$  given by  $f(\underline{x}) = \underbrace{A}_{2 \times 2} \underline{x}$

1) Superposition

$$\begin{aligned} f(\underline{x}_1 + \underline{x}_2) &= A(\underline{x}_1 + \underline{x}_2) \\ &= A\underline{x}_1 + A\underline{x}_2 = f(\underline{x}_1) + f(\underline{x}_2) \\ &= f(\underline{x}_1) + f(\underline{x}_2) \end{aligned}$$

2) Homogeneity

$$f(\alpha \underline{x}) = A(\alpha \underline{x}) = \alpha A \underline{x} = \alpha f(\underline{x})$$

$$f: \mathbb{R}^3 \rightarrow \mathbb{R}^2$$

$$f(\underline{x})$$

$$\underbrace{\left( \begin{array}{c} \phantom{0} \\ \phantom{0} \\ \phantom{0} \end{array} \right)}_{2 \times 3} \left( \begin{array}{c} x_1 \\ x_2 \\ x_3 \end{array} \right)$$

Lemma:

In general, a map

$$f: \mathbb{R}^n \rightarrow \mathbb{R}^m \text{ given by } f(\underline{x}) = \underbrace{A}_{m \times n} \underline{x}$$

is linear.

$m=n$  :  $A$  is square

$m>n$  :  $A$  is tall / thin

$m<n$  :  $A$  is wide / fat

Affine function: We say that  $f$  is affine if

$f(\underline{x}) = \underline{b} + \text{linear function}$

$$A\underline{x} \leftarrow \boxed{A} \leftarrow \underline{x}$$

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\underline{x} \leftarrow \boxed{I} \leftarrow \underline{x}$$

$$S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Swap matrix

$$S^2 = I.$$

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} x_2 \\ x_1 \end{pmatrix}$$

"matrix is what it does"

Rotation:

$$R(\theta) = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}$$

$$\underbrace{\left[ R\left(\frac{\pi}{12}\right) \right]}^{12} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$$

Recap: Functions (domains, codomains)

$$\log \sin x$$

$$(0, \pi)$$

multivariable  
fns.  $f(\underline{x}, y)$

Vectors:

$$\mathbb{R}^n$$

$$\underline{x} \in \mathbb{R}^n$$

$$\underline{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$$

$$x_1, x_2, \dots, x_n \in \mathbb{R}.$$

Linear functions

$f: \mathbb{R}^n \rightarrow \mathbb{R}^m$  is linear if

(i) Superposition

$$f(\underline{x} + \underline{y}) = f(\underline{x}) + f(\underline{y}) \quad \forall \underline{x}, \underline{y} \in \mathbb{R}^n.$$

(ii) Homogeneity

$$f(d\underline{x}) = d f(\underline{x}) \quad \forall \underline{x} \in \mathbb{R}^n, d \in \mathbb{R}.$$

Alternate def<sup>n</sup>:

$f: \mathbb{R}^n \rightarrow \mathbb{R}^m$  is linear if

$$f(d\underline{x} + \beta \underline{y}) = d f(\underline{x}) + \beta f(\underline{y}) \quad \forall \underline{x}, \underline{y} \in \mathbb{R}^n, d, \beta \in \mathbb{R}.$$

Optional: Show that these two def<sup>s</sup> are equivalent.

$f$  satisfies Def<sup>n</sup> 1  $\Rightarrow f$  satisfies Def<sup>n</sup> 2 : exercise.

$f$  satisfies Def<sup>n</sup> 2  $\Rightarrow f$  satisfies Def<sup>n</sup> 1 :  $\alpha=1 \quad \beta=1$   
 $\beta=0$

Matrix functions:

$$\underbrace{A}_{\substack{m \times n \\ \text{matrix}}} \underbrace{\underline{x}}_{\substack{\mathbb{R}^n: n \times 1 \text{ vector}}} : \underline{A\underline{x}} \in \mathbb{R}^m$$

Theorem: Any linear function  $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$  is a matrix function, i.e.  $f(\underline{x}) = A\underline{x}$  for some  $m \times n$  matrix  $A$ .

Examples:

$$\begin{array}{l} \text{Example 1: } x_i: \text{score in exam } i \\ f(x_1, x_2, \dots, x_n) = \sum_{i=1}^n w_i x_i = \underbrace{\underline{w}^T}_{\substack{\text{weight of the } i\text{th exam}}} \underline{x} \leftarrow \text{Inner product} \end{array} \quad \left| \quad \begin{array}{l} f(x, y) = x + y \\ f: \mathbb{R}^2 \rightarrow \mathbb{R} \\ f\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} \end{array} \right.$$

$$= \underbrace{(w_1 \ w_2 \ \dots \ w_n)}_{\underline{w}^T} \underbrace{\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}}_{\underline{x}}$$

$$\underline{w} = \begin{pmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{pmatrix}$$

$$\text{Input } \underline{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$$



Proof:

$$\underline{y} = A \underline{x}$$

$$\text{If } y_1 = 0, y_2 = 0 \quad y_1 \frac{(x-x_2)(x-x_3)}{(x_1-x_2)(x_1-x_3)}$$

+

$$\text{If } y_1 = 0, y_3 = 0 \quad y_2 \frac{(x-x_1)(x-x_3)}{(x_2-x_1)(x_2-x_3)}$$

+

$$\text{If } y_1 = 0, y_2 = 0 \quad y_3 \frac{(x-x_1)(x-x_2)}{(x_3-x_1)(x_3-x_2)}$$

Digression  
f: degree 2  
polynomial

$$f(x_1) = y_1 \checkmark$$

$$f(x_2) = y_2 \checkmark$$

$$f(x_3) = y_3 \checkmark$$

In general for arbitrary  $y_1, y_2, y_3$

$f(x) =$  sum of all three terms.

$$\text{Let } \underline{e}_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \begin{matrix} \leftarrow x_1 \\ \leftarrow x_2 \\ \vdots \end{matrix} \quad \underline{e}_2 = \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix} \quad \dots \quad \underline{e}_n = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}$$

$$\underbrace{\begin{pmatrix} 0 \\ \vdots \\ 1 \end{pmatrix}}_{\underline{e}_n}$$

$$\underbrace{\begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}}_{\underline{e}_1}$$

$$\underbrace{\begin{pmatrix} 0 \\ \vdots \\ 1 \end{pmatrix}}_{\underline{e}_n}$$

"Impulse vectors"

An arbitrary vector

$$\underline{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \underbrace{x_1 \underline{e}_1 + x_2 \underline{e}_2 + \dots + x_n \underline{e}_n}_{\text{"linear combination"}}$$

Now, let  $f$  be a linear function  
 $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$

$$\begin{aligned} f(\underline{x}) &= f(x_1 \underline{e}_1 + x_2 \underline{e}_2 + \dots + x_n \underline{e}_n) \\ &\quad \underbrace{\quad}_{\mathbb{R}^n} \\ &= x_1 \underbrace{f(\underline{e}_1)}_{\in \mathbb{R}^m} + x_2 \underbrace{f(\underline{e}_2)}_{\in \mathbb{R}^m} + \dots + x_n \underbrace{f(\underline{e}_n)}_{\in \mathbb{R}^m} \\ &= \underbrace{\begin{bmatrix} | & | & & | \\ f(\underline{e}_1) & f(\underline{e}_2) & \dots & f(\underline{e}_n) \\ | & | & & | \end{bmatrix}}_A \underbrace{\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}}_{\underline{x}} \end{aligned}$$

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = x_1 \begin{bmatrix} a_{11} \\ a_{21} \end{bmatrix} + x_2 \begin{bmatrix} a_{12} \\ a_{22} \end{bmatrix}$$

Another perspective on matrix mult.

① (old) Entry  $i$  in  $A\underline{x}$  is inner product (dot product)



of row  $i$  of  $A$  with  $\underline{x}$

②  $A\underline{x}$  is a weighted combination of columns of  $A$  with weights taken from  $\underline{x}$ .

$$\begin{bmatrix} 1 & 0.5 \\ 0.5 & 1 \end{bmatrix} \begin{bmatrix} 10 \\ 20 \end{bmatrix} = 10 \begin{pmatrix} 1 \\ 0.5 \end{pmatrix} + 20 \begin{pmatrix} 0.5 \\ 1 \end{pmatrix}$$

$$f(x, y) = x + y$$

$$f: \mathbb{R}^2 \rightarrow \mathbb{R}$$

$$e_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad e_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$f(e_1) = 1 \quad f(e_2) = 1$$

$$\begin{bmatrix} 1 & 1 \end{bmatrix}$$

Summary: Any  $\underline{x} \in \mathbb{R}^n$  can be written "in terms of"  
 $\underline{e}_1, \underline{e}_2, \dots, \underline{e}_n$ . And  $A\underline{e}_1 \leftrightarrow$  first column of  $A$   
 $A\underline{e}_2 \leftrightarrow$  second column of  $A$ .

Other examples of linear functions:

$$f'(t) \leftarrow \boxed{D} \leftarrow \overset{\text{not } \mathbb{R}^n}{f(t)}$$

$$g'(t) \leftarrow \boxed{D} \leftarrow g(t)$$

Exercise: Does this satisfy  
superposition & homogeneity?

Linear

## A Systems of equations:

$$2x_1 + 5x_2 + 10x_3 = 0$$

$$3x_1 + 6x_2 + 10x_3 = 5$$

$$\underset{m \times n}{A} \underset{\mathbb{R}^n}{x} = \underset{\mathbb{R}^m}{b}$$

$$\underline{b} \leftarrow \boxed{A} \leftarrow \underline{x}$$

## Solving a system of eq's

Given  $A, \underline{b}$  Find all  $\underline{x}$  such that  $A\underline{x} = \underline{b}$ .

inverse problem  
signal processing

$\{A \text{ wide} : \text{underdetermined system of equations.}$

$$A: 2 \times 100$$

$\{A \text{ tall} : \text{overdetermined system of equations.}$

machine  
learning

## Questions

① why are there infinitely many solutions to underdetermined systems?

② Is there solution to overdetermined systems?

③ When is a matrix function one-one? onto?

Suppose  $A$  is one-one.  
 $\xrightarrow{m \times n}$

$$f(x, y) = x + y \\ \mathbb{R}^2 \rightarrow \mathbb{R}$$

$$\underline{Ax_1} = \underline{Ax_2} \Rightarrow \underline{x_1} = \underline{x_2}$$

$f(x_1) = f(x_2)$

$$A(\underline{x_1} - \underline{x_2}) = 0 \Rightarrow \underline{x_1} - \underline{x_2} = 0$$

$$A\underline{x} = \underline{0} \Rightarrow \underline{x} = \underline{0}.$$

$$\underline{0} \leftarrow \boxed{A} \leftarrow \underline{x}$$

Null "space" of a matrix:

$N(A) = \{ \underline{x} \mid A\underline{x} = \underline{0} \}$  set of all inputs that return a null output.

If  $A$  is one-one then  $N(A) = \{ \underline{0} \}$   
(trivial null space)

$$A \text{ one-one} \Rightarrow N(A) = \{ \underline{0} \}$$

$$N(A) = \{ \underline{0} \} \Rightarrow A \text{ one-one?}$$

Suppose  $A$  is not one-one. then we have  $\underline{v} \neq \underline{w}$

$$\underline{v} \neq \underline{w} \text{ such that } A\underline{v} = A\underline{w}.$$

$$\text{then } A(\underline{v} - \underline{w}) = \underline{0}$$

$$\Rightarrow \underline{v} - \underline{w} \in N(A).$$

$$\underline{0} \leftarrow \boxed{\phantom{A}} \leftarrow \underline{v} - \underline{w}$$

$$A \text{ one-one} \iff N(A) = \{ \underline{0} \}.$$

the only input giving zero output must be  $\underline{0}$ .

To decide if  $A$  is one-one.

Solve  $A\underline{x} = \underline{0}$  to see if a non-zero solution  $\underline{x}$  exists.

Theorem: Any system  $A\underline{x} = \underline{0}$  with  $A$  wide ( $m < n$ )

always has a non-zero solution. In fact, such a system has infinitely many solutions.

Corollary: A wide matrix can never be one-one.

Proof:

Case:

$$w_1 = 0 \quad w_2 = 0$$

$$x_1 = *, \quad x_2 = *$$

$(w_1, w_2)$  are both not zero,

$$\boxed{\begin{matrix} m=1 \\ n=2 \end{matrix}}$$

$$\underbrace{\begin{pmatrix} w_1 & w_2 \end{pmatrix}}_A \underbrace{\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}}_{\underline{x}} = \underline{0}$$

$$\begin{aligned} x_1 &= -w_2 \\ x_2 &= w_1 \end{aligned}$$

Induction

$$\begin{pmatrix} 1 & 0 & 0 \\ * & w_{22} & w_{23} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \underline{0}$$

$$w_{22}x_2 + w_{23}x_3 = 0$$

$$\begin{matrix} m=2 \\ n=3 \end{matrix}$$

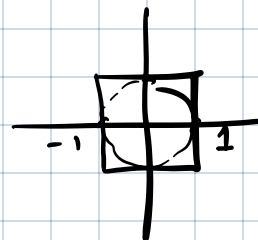
$$\begin{pmatrix} \cancel{w_{11}} & w_{12} & w_{13} \\ \cancel{w_{21}} & w_{22} & w_{23} \\ 0 & & \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \underline{0}$$

$$\begin{pmatrix} w_{12} & w_{13} \\ w_{22} & w_{23} \end{pmatrix} \begin{pmatrix} x_2 \\ x_3 \end{pmatrix} = \underline{0}$$

Thought exp

Consider an  $n$ -dimensional box

Inscribe an  $n$ -dimensional ball,



How much space in

the box  
is empty).

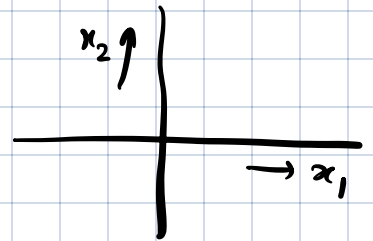
Theorem: For any wide  $A$  ( $m \times n$ ) with  $m < n$ , the system of equations  $Ax = 0$  has at least one non-zero solution.

rows: eq's  
Col's: variables

Case  $m=1, n=2$  :  $w_1x_1 + w_2x_2 = 0$

$m=2, n=3$

Approach 1:



$$\underbrace{\begin{bmatrix} \text{red oval} & \text{red circle} & \text{red circle} \end{bmatrix}}_{A \quad 2 \times 3} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Approach 2:

Assume:

$$\begin{matrix} \rightarrow \\ \rightarrow \end{matrix} \begin{bmatrix} 1 & \underline{a}^T \\ 0 & \underline{b}^T \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$m=2, n=3$

$\underline{a}$ :  $2 \times 1$

Decoupling

$$x_1 + \underline{a}^T \begin{pmatrix} x_2 \\ x_3 \end{pmatrix} = 0 \quad \text{--- (1)}$$

$$\underline{b}^T \begin{pmatrix} x_2 \\ x_3 \end{pmatrix} = 0 \quad \text{--- (2)}$$

Solve  $\underline{b}^T \begin{pmatrix} x_2 \\ x_3 \end{pmatrix} = 0$  to obtain non-zero  $(x_2, x_3)$ .

Substitute in (1) to find  $x_1$ .

In general,

$$R_1 \rightarrow \begin{bmatrix} w_1 & a_1 & a_2 \\ w_2 & b_1 & b_2 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad \text{row operation}$$

$$R_2 \rightarrow R_2 - \frac{w_2 R_1}{w_1}$$

$$\begin{bmatrix} w_1 & a_1 & a_2 \\ * & b_1 & b_2 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{cases} x_1 + x_2 = 0 \\ -x_1 - x_2 = 0 \end{cases}$$

$$\underbrace{0 = 0}$$

## Row operations

1) row multiplication

$$R_i \rightarrow \lambda R_i \quad (\lambda \neq 0)$$

(reverse)  $R_i \rightarrow R_i / \lambda$   $A \underline{x} = \underline{0}$

$$\tilde{A} \underline{x} = \underline{0}$$

$$\tilde{A} = \underbrace{\begin{bmatrix} 1 & & & 0 \\ & 1 & & \\ & & \ddots & \\ & & & \alpha \\ & & & & 1 \\ & 0 & & & & \end{bmatrix}}_{\text{diagonal matrix}} A$$

2)  $R_j \rightarrow R_j - \lambda R_i$   
 $j=2, i=1$

$j \neq i$

$$A \underline{x} = \underline{0}$$

$$\tilde{A} \underline{x} = \underline{0}$$

$$\tilde{A} = \begin{bmatrix} 1 & 0 & - & - & 0 \\ -2 & 1 & - & - & 0 \\ 0 & 0 & 1 & - & 0 \\ 0 & 0 & - & - & 1 \end{bmatrix} A$$

Exercise : check with an example system of

equations.

(reverse)  $R_j \xrightarrow{+2R_i} R_j + 2R_i$

3)  $R_i \leftrightarrow R_j$  swap  $i^{\text{th}}$  &  $j^{\text{th}}$  rows

Exercise: what is the matrix corresponding to this row operation?

Theorem: Any system  $A\underline{x} = 0$   $A (m \times n, m < n)$  has infinitely many solutions / at least one non-zero solution.

Proof: Suppose  $A: m \times n$  with first column  $(m, n > 1)$

$$\hookrightarrow \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_m \end{bmatrix}$$

Assume the first column is zero. then set  $x_1 = 1, x_j = 0$  for  $j=2, \dots, n$ . this provides a non-zero solution. Hence, we assume first column is non-zero.

Assume  $w_1 \neq 0$ . If  $w_1 = 0$  then swap  $R_1$  with  $R_j$  where  $w_j \neq 0$ .

By doing row operations

$$R_j \rightarrow R_j - \frac{w_j R_1}{w_1} \text{ for } j=2, 3, \dots, m$$

reduce the system to  $\tilde{A}\underline{x} = 0$ , where

$$\tilde{A} = \left[ \begin{array}{c|c} \begin{matrix} w_1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{matrix} & \begin{matrix} -a^T \\ B \end{matrix} \end{array} \right]$$

$m-1 \times n-1$

We know that any solution to  $\tilde{A}x = 0$  is a solution to  $Ax = 0$ .

$$\tilde{A}x = 0 \Rightarrow \begin{cases} w_1 x_1 + \underline{a}^T \begin{pmatrix} x_2 \\ \vdots \\ x_n \end{pmatrix} = 0 & \text{--- (1)} \\ \underbrace{B \begin{pmatrix} x_2 \\ \vdots \\ x_n \end{pmatrix}}_{(n-1) \times (n-1)} = 0 \end{cases}$$

By induction hypothesis, there exists a non-zero solution to  $B \begin{pmatrix} x_2 \\ \vdots \\ x_n \end{pmatrix} = 0$ . Use this solution to find  $x_1$  from (1).  
We now have a non-zero solution to  $\tilde{A}x = 0$  and hence  $Ax = 0$ .

As a base case, consider  $m=1$ . Then the system is of the form

$$a_1 x_1 + a_2 x_2 + \dots + a_n x_n = 0.$$

Assume at least one  $a_i$  is non-zero (otherwise we have infinitely many solutions). Say  $a_1 \neq 0$ .

$$\text{then set } x_1 = \frac{-a_2 x_2 - a_3 x_3 - \dots - a_n x_n}{a_1}$$

Recap:

Wide matrices cannot be one-one.

Wide matrices have infinite null-space.  $\hookleftarrow$



When can linear systems be onto?

$$\mathbb{R}^m \leftarrow \boxed{A}_{m \times n} \leftarrow \underline{x} \in \mathbb{R}^n$$

A is onto if  $\forall \underline{y} \in \mathbb{R}^m, \exists \underline{x} \in \mathbb{R}^n$  such that

$$\underline{y} = A \underline{x}.$$

Theorem:

Tall matrices cannot be onto. (dual version of earlier result).

$$\mathbb{R}^5 \leftarrow \begin{bmatrix} | & | & | \\ | & | & | \\ | & | & | \end{bmatrix}_{5 \times 2} \leftarrow \mathbb{R}^2$$

Proof: matrix tall  $\Rightarrow$  not onto ( $m=5$   
 ~~$n=2$~~ )

Assume A is onto. Then  $\exists$  vectors  $\underline{x}_1, \underline{x}_2, \dots, \underline{x}_m$

such that

$$A \underline{x}_1 = \underline{e}_1, \quad A \underline{x}_2 = \underline{e}_2, \quad \dots \quad A \underline{x}_m = \underline{e}_m$$

$$\underbrace{A}_{m \times n} \underbrace{\begin{bmatrix} \underline{x}_1 & \underline{x}_2 & \dots & \underline{x}_m \end{bmatrix}}_X = \underbrace{\begin{bmatrix} \underline{e}_1 & \underline{e}_2 & \dots & \underline{e}_m \end{bmatrix}}_{I_{m \times m}}$$

$$\underbrace{A}_{\text{tall}} \underbrace{X}_{\text{fat}} = I \quad \text{--- ①}$$

Take  $\underline{w}$  such that  $X \underline{w} = \underline{0}$

$$A X \underline{w} = \underline{0}$$

But from ①,  $A X \underline{w} = I \underline{w} = \underline{w}$

A:  
left inverse

$$BA = I$$

right inverse

$$AB = I$$

① cannot co-exist with non-trivial null space.

Hence  $A$  cannot be onto.

Exercise: Show that there cannot be more than  $n+1$  perpendicular vectors in  $\mathbb{R}^n$ . (each pair of vectors is perpendicular)

we say vectors  $\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$  are perpendicular

if  $\underbrace{x_1 y_1 + x_2 y_2 + \dots + x_n y_n = 0}$ .

- tall matrices cannot have right inverse

- wide matrices cannot have left inverse.

cannot happen.

can happen

$$\begin{bmatrix} \phantom{0} \\ \phantom{0} \end{bmatrix} \begin{bmatrix} \phantom{0} \\ \phantom{0} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \phantom{0} \\ \phantom{0} \end{bmatrix} \neq \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} 1 \end{bmatrix}$$

Exercise: Assume Tall matrices cannot be onto.  
Prove that wide matrices cannot be one-one.

Tall matrix  $\swarrow$  one-one  
 $\searrow$  ~~onto~~

$$\mathbb{R}^m \leftarrow \boxed{\phantom{0}} \leftarrow \mathbb{R}^n$$

What ensures a full matrix is one-one?

$$A = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_m \end{bmatrix}_{m \times 1}$$

$$\underbrace{A}_{m \times 1} \underbrace{x_1}_{1 \times 1} = \underbrace{A}_{m \times 1} \underbrace{x_2}_{1 \times 1}$$

$$\begin{bmatrix} a_1 x \\ a_2 x \\ \vdots \\ a_m x \end{bmatrix}_{Ax} = \begin{bmatrix} a_1 y \\ a_2 y \\ \vdots \\ a_m y \end{bmatrix}_{Ay}$$

$A_{m \times 1}$  is one-one as long as the column is non-zero.

$$\begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \\ \vdots & \vdots \\ a_m & b_m \end{bmatrix}_{m, n=2}$$

$$\begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \\ a_3 & b_3 \end{bmatrix}$$

$$\begin{bmatrix} a_1 & 0 \\ a_2 & 0 \\ a_3 & 0 \end{bmatrix} \begin{bmatrix} x \\ * \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Suppose  $A$  is one-one. Let  $B$  be a matrix obtained by rearranging the columns of  $A$ . Then

$B$  one-one  $\Rightarrow A$  is one-one.

one-one men seems to be a property of a collection of vectors.

Linear independence: We say that a collection of vectors  $\underline{a}_1, \underline{a}_2, \dots$  is linearly independent (LI) if

$$A = [\underline{a}_1 \quad \underline{a}_2 \quad \dots \quad \underline{a}_n] \text{ is } \underline{\text{one-one}}.$$

$\hookrightarrow \text{null}(A) \text{ is } \{\underline{0}\}.$

Since  $A$  is one-one.

$$\underline{A}\underline{x} = \underline{0} \Rightarrow \underline{x} = \underline{0}.$$

$$[\underline{a}_1 \quad \underline{a}_2 \quad \dots \quad \underline{a}_n] \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \underline{0} \Rightarrow x_1 = x_2 = \dots = x_n = 0$$

$$x_1 \underline{a}_1 + x_2 \underline{a}_2 + \dots + x_n \underline{a}_n = \underline{0} \Rightarrow x_1 = x_2 = \dots = x_n = 0.$$

$$\underline{a}_1 = \frac{-x_2 \underline{a}_2 - x_3 \underline{a}_3 - \dots - x_n \underline{a}_n}{x_1}$$

$$a_1 = -a_2 - 30a_3$$

$$x_1 = 1, x_2 = 1, x_3 = 30, x_4 = x_5 = \dots = 0.$$

If  $\underline{a}_1, \underline{a}_2, \dots, \underline{a}_n$  are LI, we cannot express any one of the  $\underline{a}_i$ 's in terms of the others.

Theorem: 1) There can be at most  $n$  linearly independent vectors in  $\mathbb{R}^n$ .

2) Any collection of  $n+1$  vectors in  $\mathbb{R}^n$  is linearly dependent.

Proof: Construct the matrix with these vectors as columns.  
 this matrix is  $n \times (n+1)$  i.e. wide. Hence it cannot be  
 one-one. By definition, the columns cannot be linearly  
 independent.

linear independence  $\iff$  degrees of freedom.  
 $=$



Recap: functions domains/codomains  
 one-one onto

linear functions from  $\underbrace{\mathbb{R}^n}_{\text{domain}}$  to  $\underbrace{\mathbb{R}^m}_{\text{codomain}}$ .

matrix mult.

$\mathbb{C}^n$   
 $\{0,1\}^n$

$m \times n$   $m < n$   
 $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$

$$\begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$$

$\leftarrow \boxed{A} \leftarrow$

null space

$$n(A) = \{ \underline{x} \mid A\underline{x} = \underline{0} \}$$

wide matrices cannot be one-one, can be onto.

tall matrices cannot be onto, but can be one-one

Suppose  $A$  is  $m \times n$ , with  $m \geq n$ . When is  $A$  one-one?  
 this happens only if  $n(A) = \{ \underline{0} \}$ .

$$A\underline{x} = \underline{0} \implies \underline{x} = \underline{0}$$

Let  $\underline{a}_1 \quad \underline{a}_2 \quad \underline{a}_3 \quad \dots$  be the columns of  $A$

$$A\underline{x} = \begin{bmatrix} \underline{a}_1 & \underline{a}_2 & \underline{a}_3 & \dots \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \dots \end{bmatrix}$$

one-one

$$A\underline{x} = \underline{0} = A\underline{0}$$

$$\implies \underline{x} = \underline{0}$$

$$n(A) = \{ \underline{0} \}$$

$$A\underline{x}_1 = A\underline{x}_2, \text{ if } n$$

$$\text{then } A(\underline{x}_1 - \underline{x}_2) = \underline{0}$$

$$\underbrace{\quad\quad\quad}_A \begin{bmatrix} \underline{a_1} \\ \underline{a_2} \\ \vdots \\ \underline{a_n} \end{bmatrix} = x_1 \underline{a_1} + x_2 \underline{a_2} + \dots + x_n \underline{a_n} \quad ] \text{ linear combination.}$$

Say,  $x_1 = 0.5 \neq 0$

$$\frac{x_1 \underline{a_1} + x_2 \underline{a_2} + \dots + x_n \underline{a_n}}{x_1} = 0$$

$$\underline{a_1} = -\frac{x_2}{x_1} \underline{a_2} - \frac{x_3}{x_1} \underline{a_3} - \dots$$

Linear independence: is a property of a collection of vectors.

Let  $\underline{a_1}, \underline{a_2}, \dots, \underline{a_n}$  be the vectors and

$A = [\underline{a_1} \ \underline{a_2} \ \dots \ \underline{a_n}]$  be the corresponding matrix.

the following statements are equivalent

(i)  $A$  is one-one

(ii)  $\text{null}(A) = \{ \underline{0} \}$

(iii)  $\underline{a_1}, \underline{a_2}, \dots, \underline{a_n}$  are linearly independent

(iv) None of the vectors  $\underline{a_i}$  can be written as a linear combination of the other vectors.

Linear dependence: Let  $\underline{a_1}, \underline{a_2}, \dots, \underline{a_n}$  be a collection of

vectors,  $A = [\underline{a_1} \ \underline{a_2} \ \dots \ \underline{a_n}]$ .

the following are equivalent

(i)  $A$  is not one-one

(ii)  $\text{null}(A) \supset \{0\}$

(iii)  $\underline{a}_1, \underline{a}_2, \dots, \underline{a}_n$  are linearly dependent

(iv) One of the  $\underline{a}_i$  can be written as a linear combination of some of the other  $\underline{a}_i$ .

Example

①  $\{\underline{e}_1, \underline{e}_2, \underline{e}_3, \dots\}$  always LI.

$$\underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ & 0 \\ 0 & 1 \end{bmatrix}}_A \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_1 \end{bmatrix}$$

②  $\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}$

$$\underbrace{\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}}_A \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 + x_2 + x_3 \\ x_2 + x_3 \\ x_3 \end{bmatrix}$$

Suppose  $\underline{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ , let  $A\underline{x} = 0$

then from above,  $x_1 + x_2 + x_3 = x_2 + x_3 = x_3 = 0$

which is only possible when  $x_1 = x_2 = x_3 = 0$ .

hence  $\text{null}(A)$  includes only  $\underline{0}$ .

③  $\left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} \right\}$

Exercise:

$$④ \quad \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \right\} \quad \text{im}(A) = \left\{ \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\}$$

linearly dependent

$$\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} = \underline{0 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + 0 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}}$$

$$⑤ \quad \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ -1 \end{bmatrix} \right\}$$

Properties: Any subset of a linearly independent set is linearly independent.

Any superset of a linearly dependent set is linearly dependent.

Proof:

Define:  $\text{im}(A)$  read image space of  $A$   $\text{im}(A)$ .  
range of  $A$

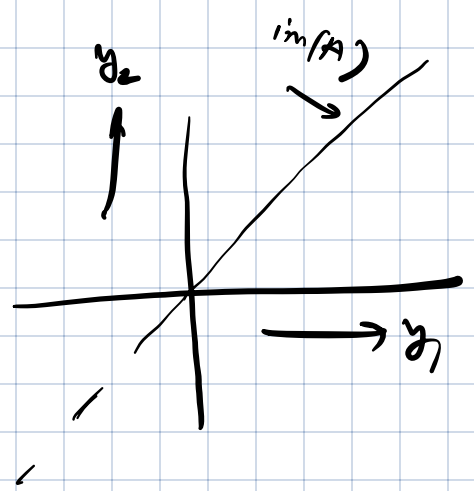
$$\begin{aligned} \text{im}(A) &= \left\{ \underline{y} \in \mathbb{R}^m : \underline{y} = A\underline{x} \text{ for some } \underline{x} \in \mathbb{R}^n \right\} \\ &= \left\{ A\underline{x} : \underline{x} \in \mathbb{R}^n \right\} \end{aligned}$$

$$A = \begin{bmatrix} 1 & 5 \\ 1 & 5 \end{bmatrix} \quad m=n=2$$



$$\text{im}(A) = \left\{ \begin{bmatrix} 1 & 5 \\ 1 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right\}$$

$$= \left\{ \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} x_1 + 5x_2 \\ x_1 + 5x_2 \end{pmatrix} \right\}$$



$$A = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \quad m, n = 2.$$

$$\text{im}(A) = \left\{ \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} x_1 + x_2 \\ x_1 - x_2 \end{pmatrix} \right\}$$

$$\text{im}(A) = \mathbb{R}^2.$$

$$A = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\text{im}(A) = \left\{ \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right\}.$$